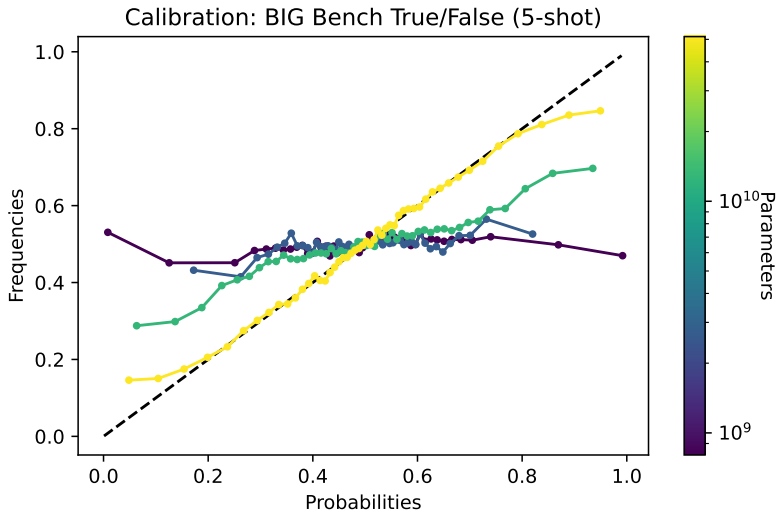


Figure 8 We show calibration curves for various model sizes on all of the multiple choice tasks in BIG Bench, reformulated as True/False questions on a mix of the correct answers, and randomly chosen incorrect answer options. The 52B model is very well-calibrated except near the tails, where it is overconfident.

```
Question: Who was the first president of the United States?
Choices:
 (A) Barack Obama
 (B) George Washington
 (C) none of the above
Answer:
```

We found that this procedure degraded performance very significantly on evaluations; results for MMLU are shown in Figure 36 in the appendix. Furthermore, adding "none of the above" also harms calibration, as can be seen in Figures 5 and 7. It seems that even the 52B model is biased against using the “none of the above” option and failed to use it with appropriate frequency. This is particularly surprising for 5-shot evaluations; we also tried evaluating 20-shot and this also did not improve performance.

3.2 Models are Well-Calibrated on True/False Tasks

We saw in section 3.1 that language models seem to be confused by a “none of the above” option. Here we take a different approach, and simply ask models if a given answer is true or false. So we use the format:

```
Question: Who was the first president of the United States?
Proposed Answer: George Washington
Is the proposed answer:
 (A) True
 (B) False
The proposed answer is:
```

where we expect either ‘ (A)’ or ‘ (B)’ as an answer. If the model responses are correct at more than chance level, and especially if they are calibrated, then the probability P(True) indicates whether the model believes a response is valid.

As a first test of this approach, we can use answer options from existing multiple choice tasks. For this purpose, we take the correct answer and a randomly chosen incorrect answer, and create a new evaluation set with twice as many problems in the format above, asking models to determine if each answer is correct. In Figure 8 we show the calibration results from this method on BIG Bench. We see that the 52B model is quite well-calibrated in this context. We show similar results for MMLU in Figure 36 in the appendix, and scaling trends comparing a variety of evaluation methods in Figure 7.

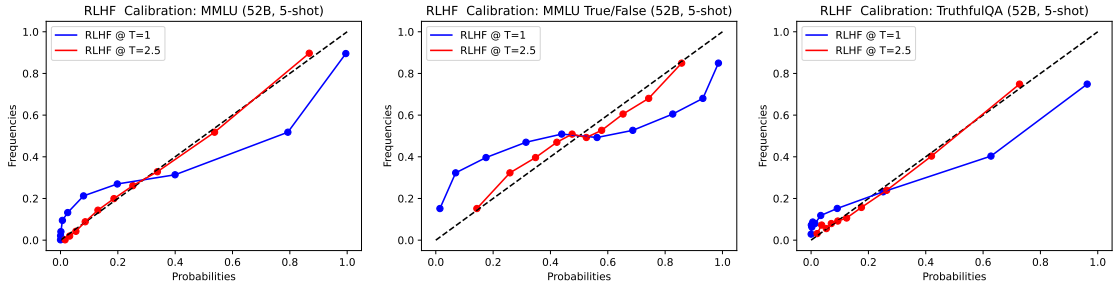


Figure 9 We show calibration curves for RLHF policies finetuned from our language models. Calibration of these models appears to be very poor, but simply adjusting the temperature of their probability distributions to $T = 2.5$ largely fixes calibration issues for three different evaluations.

3.3 RLHF Policy Miscalibration Can Be Remediated with a Temperature Tuning

Our focus in this paper is on pure language models, but as a quick experiment we also looked at calibration for a helpful and harmless RLHF policy, trained exactly as in [Bai et al., 2022] using the base language models we are studying here. We find that these policies naively appear very miscalibrated, which is not surprising, since RL finetuning tends to collapse language model predictions towards behaviors that receive the most reward. However, a simple temperature adjustment (with the same temperature $T = 2.5$ for all evaluations) largely fixes calibration issues with several independent evaluation tasks, with results shown in Figure 9. This also matches what was found for gender bias evaluations in [Bai et al., 2022], where RLHF policies had a bias score very similar to a basic language model evaluated at lower temperature. Of course more intensive RL training might distort calibration in ways that cannot be remedied in this way, but this at least provides hope that good calibration can survive some forms of finetuning.

4 Ask the AI: Is your proposed answer True or False?

We will now apply the True/False approach from section 3.2 to the samples models generated when trying to answer questions, including the short answer tasks arithmetic, Lambada, and TriviaQA, and the long-form answer tasks Codex HumanEval and GSM8k (technically GSM8k calls for a short answer, but we will be evaluating full written solution attempts, which have been solicited via chain-of-thought/scratchpad [Nye et al., 2021, Wei et al., 2022] prompting). We show the accuracies when sampling at a temperature of one in Figure 1.

In almost all cases self-evaluation performance improves with model size, and for our 52B models answers labeled with P(True) > 50% are far more likely to be correct as compared to generic responses (see the summary histogram and comparisons in Figure 1). We also find that showing the model many $T = 1$ samples for a single question significantly improves its ability to evaluate whether any given sample is correct. This is somewhat reminiscent of self-consistency [Wang et al., 2022], though here we are showing models their own samples, rather than externally judging self-consistency ourselves. One way to summarize these results is via the Brier scores for self-evaluation, which are shown in Figure 11.

We provide complete results in appendix C; since there are quite a few figures we only show representative examples from Lambada and Codex in the main text.

4.1 Basic Self-Evaluation

In section 3.2 we saw that large language models are well-calibrated on True/False questions. Our primary motivation for studying this issue was to ask language models about their own outputs. That is, we are interested in a process where we first ask a question like:

```
Question: Who was the first president of the United States?
Answer:
```

and sample a response from the model. Then we ask the model about its own sample:

```
Question: Who was the first president of the United States?
Proposed Answer: George Washington was the first president.
```



Figure 10 Models self-evaluate their own samples by producing a probability P(True) that the samples are in fact correct. Here we show histograms of P(True) for the correct and incorrect samples, in the evaluation paradigm where models also see five $T = 1$ samples for the same question, in order to improve their judgment. Here we show results only for Lambada and Codex, as these are fairly representative of short-answer and long-answer behavior; for full results see Figure 28 in the appendix.

```
Is the proposed answer:
 (A) True
 (B) False
The proposed answer is:
```

and evaluate the probabilities it assigns to the ' (A)' and ' (B)' options. We apply this approach to model generated answers from TriviaQA [Joshi et al., 2017], Lambada [Paperno et al., 2016], the Codex HumanEval [Chen et al., 2021], GSM8k [Cobbe et al., 2021], and basic arithmetic problems.

This True/False self-evaluation may be considerably more difficult than the tasks we studied in section 3.2, because there the model was presented with human-written possibilities, whereas here the model is forced to evaluate[4] its own samples. These samples may lie close to the model's own decision boundary for validity (or the model may simply be overconfident about its own samples), whereas options presented to the model by a third party may be easier to categorize. We observe this effect in several ways. Zero-shot, P(True) is poorly calibrated, and typically it lies close to 50% for typical samples (see Figure 38 in the appendix for illustrations). Furthermore, we show in Figure 33 in the appendix that samples from smaller models are universally easier to categorize as correct or incorrect, for models of all sizes.

4.2 Showing Many $T = 1$ Samples Improves Self-Evaluation

We can improve performance further by showing the model other $T = 1$ samples, for comparison. That is, we generate 5 samples in total, and then ask the model about the validity of one of these samples:

```
Question: Who was the third president of the United States?
Here are some brainstormed ideas: James Monroe
Thomas Jefferson
John Adams
Thomas Jefferson
George Washington
Possible Answer: James Monroe
Is the possible answer:
 (A) True
 (B) False
The possible answer is:
```

With this format, performance improves significantly on all of the short-form answer tasks, as compared to the version where we only show models a single proposed answer, as summarized in Figure 11. However,

[4]Language models do not typically have an explicit way of knowing if tokens were provided to them by a third party, or are their own samples. This may make self-evaluation more difficult (or at least ambiguous), and can have other issues in an RL setting [Ortega et al., 2021].